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QUESTION BANK

Semester – III Examinatation – 2026

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MAJ-4

ARTIFICAL INTELLIGENCE

FUNDAMENTALS OF ARTIFICIAL INATELLIGENCE

UNIT I – Overview of AI, Problems, Problem Space and Search

2 Marks

1. Define Artificial Intelligence (AI).
2. Mention any two applications of AI.
3. Mention any two major goals of AI.
4. Differentiate between Strong AI and Weak AI.
5. What is the Turing Test?
6. What is AI Technique?
7. What is Problem Space Search?
8. What is a Production System?
9. Mention the characteristics of a Production System.
10. What is Heuristic Search?
11. What is Hill Climbing?
12. What is Best-First Search?
13. What is Generate and Test?
14. What is Problem Reduction?
15. What is Constraint Satisfaction?

5 Marks

1. Explain the importance of AI in modern computing with real-world applications.
2. Discuss the concept of Problem Space Search with an example.
3. Explain the Best-First Search Technique with an example.
4. Define Heuristic Search. Explain the role of Heuristic Functions with suitable examples.
5. Compare Uninformed Search and Heuristic (Informed) Search Techniques. Provide advantages and limitations.
6. Explain Hill Climbing Search with an example.
7. Explain Generate and Test Technique.
8. Explain Production System and its characteristics.
9. Explain Problem Reduction with an example.
10. Explain Constraint Satisfaction Problems with examples.

UNIT II – Knowledge Representation

2 Marks

1. What is Knowledge?
2. What is Knowledge Representation?
3. Why is Knowledge Representation needed in AI?
4. What is Declarative Knowledge?
5. What is Procedural Knowledge?
6. What is Semantic Network?

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7. Mention one advantage of Semantic Networks.
8. What are Frames in AI?
9. Give one example of a Frame.
10. What is Knowledge Representation System?
11. What is Inference in Knowledge Representation?
12. What is the Frame Problem in AI?

5 Marks

1. Compare different approaches to Knowledge Representation in AI.
2. Explain the characteristics of a Representation System with examples.
3. Explain Semantic Networks and Frames. How do they help in representing structured knowledge?
4. Discuss the major issues in Knowledge Representation, including inference, consistency, completeness and expressive adequacy.
5. Explain the Frame Problem in AI and its significance with examples.
6. Differentiate between Declarative and Procedural Knowledge.
7. Explain different approaches used in Knowledge Representation.
8. Explain the advantages and limitations of Semantic Networks and Frames.

UNIT III – Using Predicate Logic

2 Marks

1. What is Predicate Logic?
2. What is Propositional Logic?
3. What is a Predicate?
4. What is a Quantifier?
5. What is Universal Quantifier (∀)?
6. What is Existential Quantifier (∃)?
7. What is a Variable in Predicate Logic?
8. What is a Constant?
9. What is a Function in Predicate Logic?
10. What is a Relation in Predicate Logic?
11. What is a Computable Function?
12. What is a Computable Predicate?
13. What is the difference between Propositional Logic and Predicate Logic?

5 Marks

1. Describe Propositional and Predicate Logic as Knowledge Representation Techniques. Highlight their differences with examples.
2. Explain the role of Predicate Logic in AI.
3. Explain how simple facts are represented in Predicate Logic with examples.
4. Explain how instances and Is-a relationships are represented in Predicate Logic.
5. Explain Universal and Existential Quantifiers with suitable examples.
6. Explain Computable Functions and Predicates with examples.
7. Differentiate between Propositional Logic and Predicate Logic.

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UNIT IV – Natural Language Processing and Learning

2 Marks

1. What is Natural Language Processing (NLP)?
2. What is Syntactic Processing?
3. What is Semantic Processing?
4. What is Discourse Processing?
5. What is Pragmatic Processing?
6. What is Machine Learning?
7. What is Rote Learning?
8. What is Learning by Taking Advice?
9. What is Learning from Examples?
10. What is Explanation-Based Learning?
11. What is Learning in Problem Solving?
12. What is Example-Induction?

5 Marks

1. What is Semantic Processing in NLP? Explain with an example.
2. Explain Learning by Taking Advice and Learning from Examples with illustrations.
3. Explain the introduction and applications of Natural Language Processing.
4. Explain Syntactic, Semantic, Discourse and Pragmatic Processing.
5. Explain Rote Learning with an example.
6. Explain Learning by Taking Advice.
7. Explain Learning from Example-Induction.
8. Explain Explanation-Based Learning.
9. Explain Learning in Problem Solving.

UNIT V – Expert System

2 Marks

1. What is an Expert System?
2. Mention any two applications of Expert Systems.
3. What is Domain-Specific Knowledge?
4. What is a Knowledge Base?
5. What is an Inference Engine?
6. What is an Expert System Shell?
7. What is LISP?
8. Mention any two features of LISP.
9. What is AI Programming Language?
10. What is the role of an Expert System in AI?

5 Marks

1. What is an Expert System? Describe its components and applications.
2. Discuss the importance of data in Machine Learning and AI.
3. Explain the role of LISP in AI Programming and discuss its key features.
4. Explain the introduction and working principles of Expert Systems.
5. Explain the role of Domain-Specific Knowledge in Expert Systems.
6. Explain Expert System Shells with examples.
7. Discuss the advantages and limitations of Expert Systems.
8. Explain the importance of LISP and other AI Programming Languages.

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